FN-CUSTOS-003

A detected regression, bounded: the action gate under a blind pre-registered injection

Prepared for	Tavares Heath — third reconstruction (run 11, sabotage 003): a detected regression on a blind, pre-registered run
Reconstruction ID	`FN-da06acd59938f5b9`
Engine version	camsva 0.9.0
Generated	2026-10-20
Status	**Reconstruction: FINAL — BOUNDED (closed against the specified admitted observation set and reconstruction rules; it does NOT establish completeness, independent production of the records, or the absence of other relevant evidence). Publication: PENDING SUBJECT REVIEW**

Purpose & boundary

This states, per proposition, what the records **establish**, **contradict**, or **cannot establish**.

STRICTLY NON-DISPOSITIVE, and this is the third experiment of a deliberate series. Roles, no overlap: CUSTOS-CORE defined and exercised the control (its pre-flight action gate) and froze the pre-registration; the workflow produced the record. The experiment was DESIGNED and REPORTED as blind: the pre-registration and the first preserved CI execution (with the DESIGN.md hash verified before any 003 step) provide OBSERVABLE protocol evidence, while the absence of local pre-execution and prior inspection of the gate's test file remains SUBJECT-DECLARED. FactNotebook reconstructs what that record supports, contradicts or cannot establish. FN did not define or enforce the control. The record establishes four things, each scoped: the verdict follows its declared rule (P1); the record contains and binds a sealed DESCRIPTION of the injected modification (P2 — documentary, with the standing reservation that a hash records a description but does not prove executed==diffed); the clean and injected probe records report DENY and ALLOW on the identical request (P3 — an OBSERVED DELTA of two recorded decisions, not a decision measured against an expectation); and the suite detected this regression through two named assertions at two levels (P5). Two things it does NOT establish. Whether that delta is a control violation CAUSALLY attributable to the injected modification is CANNOT ESTABLISH on the admitted observations (P4): the frozen record provides evidence of tracked-state equivalence (one-line diff, same CI job and commit f5d48c9, in-process tenant state) but does not hash-bind the clean probe's executed code or exclude untracked / environmental deltas, so determining-cause is a bounded inference, not an observation (the forward-instrumented run 38811246068 introduces additional per-probe instrumentation to address that residual gap, but its effect on causal establishability needs a separate reconstruction and is not determined here). Universal coverage is CANNOT ESTABLISH (P6): 30 other gate tests passed under this injection (recorded as a fact, not explained by a scope analysis), so RED-CONFIRMED is a detected regression, not a clean bill of health. Two boundaries are held explicitly. First, even IF the causal attribution (P4) were established, an ALLOW would be a violation at the authorization-DECISION boundary — the gate permitting what it should deny — NOT that the tool executed or that a real-world effect occurred; the same line the Sovereign Veritas X5 reconstruction draws between a recorded decision and a real effect. Second, 'detected' (P5) is not 'exhaustively guarded' (P6). The series arc: run 9 could not establish coverage of the regression class (pytest -x); that honest CANNOT ESTABLISH functioned as a specification; the subject improved the instrument (v3, then this blind pre-registered probe) — without FactNotebook enforcing, re-running or controlling anything. Independent reconstruction does not change the past; it can change what a system makes provable next. No Exempt Node: every observation is a record from the subject's own run, now frozen in-repo and re-hashed here against its manifest (match); FN does not exempt itself. Corroboration is same-workflow, not mutual independence of source.

Provenance & processing

Every observation points to one public CI run (38610363998 / run 11, commit f5d48c9), now frozen in-repo at docs/red-proof-records/run-38610363998/ with a SHA-256 manifest (MANIFEST.md). Every digest this reconstruction cites — the sealed injection diff (SHA-256 74456f76…), both probe artifacts (clean / injected), the

JUnit, pytest output and exit code — was re-fetched and re-hashed here and MATCHES the manifest. Matching digests show the frozen set is internally consistent with its own manifest; the files remain products of the same workflow run (same-workflow corroboration, not mutual independence of source). Publication is gated on the subject's draft review.

Scope & coverage

Records were constituted and supplied by Tavares Heath's CUSTOS-CORE red-proof CI (the SUBJECT; public repo ceyptoslim/CUSTOS-CORE, AGPL). The record is the subject's own CI output for a blind, pre-registered run; it is NOT a source independent of the subject, and admitted without selective exclusion of candidate observations — all extracted candidates were admitted. This finding is bounded to that submitted surface; the absence of internal selection does not establish the submission's completeness relative to the represented system, nor whether an adverse class of observations was withheld — that completeness is not established.

See Annex C (acquisition / selection boundary) and Annex B (non-admitted candidate register).

Results

Every tested proposition with its engine state and, where one applies, the determining limitation. Full detail follows below.

Proposition	State	Determining limitation
P1-verdict-follows-rule — the recorded verdict RED-CONFIRMED follows the declared verdict rule (pytest_exit_code != 0 -> RED-CONFIRMED) applied to the recorded exit code (1)	**ESTABLISHED**	—
P2-modification-recorded — the submitted record contains a sealed description of the intended modification, identified by a SHA-256 value and referenced by the run identity	**ESTABLISHED**	—
P3-probe-delta — the submitted clean and injected probe records report DENY and ALLOW, respectively, for an identically-specified request	**ESTABLISHED**	—
P4-violation-attribution — the injected modification is the determining cause of the ALLOW — i.e. a control violation exercised and causally attributable to this injection	**CANNOT ESTABLISH**	evidence of tracked-state equivalence (one-line diff, same job/commit f5d48c9); clean-probe code not hash-bound + untracked/env deltas not excluded -> bounded inference, not observation (forward run 38811246068 adds instrumentation for that gap — a separate reconstruction, not determined here)
P5-detection-scoped — the test suite detected this regression: the two identified admin-auth assertions failed with the injected behaviour	**ESTABLISHED**	—
P6-universal-coverage — RED-CONFIRMED establishes that the gate's authority enforcement is correct in general — that every authority-weakening regression would be detected	**CANNOT ESTABLISH**	detection of THIS regression by two assertions; 30 other gate tests passed under this injection — a detected regression, not a clean bill of health

1 — What the record establishes

Proposition	State	Basis

the recorded verdict RED-CONFIRMED follows the declared verdict rule (pytest_exit_code != 0 -> RED-CONFIRMED) applied to the recorded exit code (1)	**ESTABLISHED**	ESTABLISHED. From the record (frozen in-repo, re-hashed here against MANIFEST.md): pytest_exit_code = 1; rule 'exit_code != 0 -> RED-CONFIRMED'; verdict = RED-CONFIRMED. 1 != 0 => RED-CONFIRMED. Internal consistency of the record — same check as run-9/v3 P1. The record's digests are re-hashed here against the frozen MANIFEST.md (match)
the submitted record contains a sealed description of the intended modification, identified by a SHA-256 value and referenced by the run identity	**ESTABLISHED**	ESTABLISHED — as a DOCUMENTARY relation, not as execution. The record carries a sealed post-injection diff whose SHA-256 (74456f76…) is referenced by the run identity (injection_diff_sha256) — 'sealed' here means a SHA-256 byte-commitment bound into the record identity, not a separately-demonstrated temporal seal or independent custody; the diff describes the authority check in custos/action_gate.py weakened so an un-registered tool proceeds. What this establishes is that the record CONTAINS and BINDS a description of the intended modification — NOT that the modified code was the code executed (a hash records a description, it does not prove executed == diffed; FactNotebook does not re-run). A stronger 'the modification was executed' proposition would need an additional basis (e.g. before/after execution evidence) and is not claimed here. The 74456f76… value was re-hashed here from the frozen in-repo patch and matches the directory manifest

the submitted clean and injected probe records report DENY and ALLOW, respectively, for an identically-specified request	**ESTABLISHED**	ESTABLISHED — a documentary/observational proposition about the two probe records (both re-hashed here = match the frozen manifest). On the same request (tool 'payment_transfer', un-registered, tenant 'probe_tenant_003'), the CLEAN phase records allowed=false / deny with the rule action_gate_tool_not_allowlisted FIRED, and the INJECTED phase records allowed=true with NO rule fired (triggered_rule null, 'No policy violations detected'). The injected artifact carries run_id + injection_diff_sha256 (74456f76…); the clean artifact carries null ONLY in the injection_diff_sha256 field (both artifacts carry the same run-identity fields — the clean phase simply has no injection to reference). This clean/injected pairing is far stronger than a single post-injection test — but it is a statement about the two RECORDED decisions, not yet a causal claim about the modification (that is P4-violation-attribution)

the test suite detected this regression: the two identified admin-auth assertions failed with the injected behaviour	**ESTABLISHED**	ESTABLISHED, scoped. The full suite ran without -x: 518 cases, 2 failed, exit 1. The two failing tests are the intended assertions of this class — TestFailClosed::test_unregistered_tenant_denies_all_tools and TestEvaluateActionAPI::test_endpoint_denies_unallowlisted_tool — and both failure messages assert the injected behaviour itself (allowed=True where deny is required), at two levels (fail-closed unit + API endpoint). IDENTIFICATION CRITERION: tests asserting that an un-registered tool is denied by the gate. What is ESTABLISHED is the NARROW proposition: these two named, executed assertions failed under the injection (directly in the JUnit). This is the 'detected' counterpart to run-9, where coverage of the class was CANNOT ESTABLISH under -x

Limitations

P4-violation-attribution — *the injected modification is the determining cause of the ALLOW — i.e. a control violation exercised and causally attributable to this injection* → **CANNOT ESTABLISH** (evidence of tracked-state equivalence (one-line diff, same job/commit f5d48c9); clean-probe code not hash-bound + untracked/env deltas not excluded -> bounded inference, not observation (forward run 38811246068 adds instrumentation for that gap — a separate reconstruction, not determined here)). CANNOT ESTABLISH — the most interesting boundary of 003, and now a PRECISE one rather than an assumed gap. The frozen record carries real equivalence evidence, which this reconstruction weighed: both probe artifacts record the same run_id (38610363998) — sequential steps of one CI job, same runner workspace, same checkout of commit f5d48c9; the sealed diff enumerates the complete set of TRACKED-file changes between the two phases (one file, one line in custos/action_gate.py); and the probe constructs its tenant state in-process at that commit, so the applicable policy and state are specified by the probe script itself. That provides evidence of tracked-state equivalence. What it does NOT establish — and the frozen directory's README states this same boundary — is a hash-level binding of the CLEAN probe's executed code (the clean artifact predates the diff seal and carries no code-state hash) or the exclusion of untracked-file / environmental deltas within the job. So determining-cause is a BOUNDED INFERENCE across an enumerated one-line tracked diff within one job — strong, but not an observation. Held at CANNOT ESTABLISH; the subject explicitly accepts this. This is exactly '_what observations suffice to attribute a violation to a determined modification, independently of the test verdict?_'. FORWARD INSTRUMENTATION (not evaluated here): a subsequent run 38811246068 introduces additional per-probe instrumentation (each phase's own script SHA-256 and a `git status --porcelain` capture) intended to address this residual equivalence gap. Its effect on causal establishability requires a SEPARATE reconstruction and is not determined by this run-11 finding; run 11 stands as recorded. BOUNDARY regardless: even IF established, an ALLOW would be a violation at the authorization-DECISION boundary — the gate permitting what it should deny — NOT that the tool executed or that a real-world effect occurred (the same line drawn in the Sovereign Veritas X5 case).

P6-universal-coverage — *RED-CONFIRMED establishes that the gate's authority enforcement is correct in general — that every authority-weakening regression would be detected* → **CANNOT ESTABLISH** (detection of THIS regression by two assertions; 30 other gate tests passed under this injection — a detected regression, not a clean bill of health). CANNOT ESTABLISH — and this boundary is on the record deliberately. The detection surface for THIS class is exactly the two assertions that failed; the other 30 dedicated gate tests passed under this injection (this reconstruction records that they passed, not a scope analysis of why they did). So RED-CONFIRMED is the detection of THIS regression at those two levels — not universal coverage of every possible authority weakening, and not a clean bill of health for the gate. Establishing the universal would require classifying the whole suite's scope against the space of authority weakenings, which this record does not do. 'Detected' is not 'exhaustively guarded'

Separation of authority

Observation instruments determine what they observe and propose. Admission determines what may enter the reconstruction record. FactNotebook determines what that admitted record permits us to establish about the proposition. An instrument may search, select, transform or directly acquire observations according to its own declared procedure; those operations remain attributable properties of the instrument, not hidden properties of the reconstruction engine.

recorded behaviour → **observation** (attributed to its instrument) · declared risk / rule classifications → **the evaluator's / authority's policy** (never engine facts) · the consequential judgment → **the accountable authority**, outside this reconstruction.

Re-derivability

Two levels are preserved. **The finding** re-derives: re-running the reconstruction against the admitted observations reproduces the reconstruction identifier `FN-da06acd59938f5b9` and the states in this document — with no re-run of the probabilistic extraction. **The admitted record's boundary** is separately inspectable via the candidate admission register (Annexes A/B) and the acquisition boundary (Annex C). *Version the instrument · trace the extraction · freeze the admitted record · replay the reconstruction.*

References

Each observation points to a citable source a third party can re-walk; the status records how it was verified.

CUSTOS-CORE red-proof record — sabotage 003 (run 38610363998 / run 11) — workflow_dispatch on merged main; DESIGN.md hash verified by the workflow before any 003 step; pytest_exit_code 1, verdict RED-CONFIRMED; binds injection_diff_sha256 74456f767f1cba7c… into the record identity; references clean+injected probe decisions and the full JUnit (no -x) · *PRIMARY — frozen in-repo (003-weaken-action-gate/); re-hashed here, matches MANIFEST.md* · `https://github.com/ceyptoslim/CUSTOS-CORE/tree/main/docs/red-proof-records/run-38610363998/003-weaken-action-gate (run 38610363998)`

Pre-registration (DESIGN.md), frozen before the first run — names the target control (pre-flight action gate), the injected modification class (weaken the authority check so an un-registered tool proceeds), the expected violation criterion, and the probe definition (clean + injected, un-registered tool, known-allowlist tenant) · *PRIMARY — on main via PR before run 11; its hash was verified by the workflow before 003 ran* · `https://github.com/ceyptoslim/CUSTOS-CORE (docs/red-proof-records/sabotage-003/DESIGN.md; hash pinned and verified by the workflow)`

Control probe (clean + injected) — the observed delta — same un-registered-tool request on both builds: CLEAN -> DENY (triggered_rule action_gate_tool_not_allowlisted); INJECTED -> ALLOW (no rule fired, 'No policy violations detected'); injected probe artifact carries run_id + injection_diff_sha256 byte-matched to the record identity · *PRIMARY — frozen in-repo (003-probe/); both artifacts re-hashed here, match MANIFEST.md* · `docs/red-proof-records/run-38610363998/003-probe/`

Execution evidence (JUnit, full suite, no -x) — 2 failed, 481 passed, 35 skipped (518 cases), exit 1; failures = TestFailClosed::test_unregistered_tenant_denies_all_tools + TestEvaluateActionAPI::test_endpoint_denies_unallowlisted_tool (allowed=True where deny required); 30 other dedicated gate tests passed · *PRIMARY — frozen in-repo (003-weaken-action-gate/); re-hashed here, matches MANIFEST.md* · `docs/red-proof-records/run-38610363998/003-weaken-action-gate/`

The before/after lineage (run 9 → v3 → 003) — FN-CUSTOS-REDPROOF-001 (run 9, record v2, -x) coverage CANNOT ESTABLISH; -002 (v3, no -x) coverage ESTABLISHED; FN-CUSTOS-003 (run 11) a DIFFERENT control (action gate) on a blind pre-registered run, DETECTED, with an observed-delta probe; the determining-cause CANNOT ESTABLISH of run 11 then specified the forward instrumentation shipped in run 38811246068 (per-probe script-SHA + git-status). The CANNOT ESTABLISH of run 9 functioned as a specification; the subject improved the instrument; FN never touched the system ·

REFERENCE — the trio · `https://github.com/ceyptoslim/CUSTOS-CORE/tree/main/docs/red-proof-records`

Observation register (obs -> provenance -> integrity -> probative role) — Observation count (4) is not proposition count (6). P1 <- record verdict/exit (A2, internal consistency). P2 <- sealed diff referenced by identity (A2, DOCUMENTARY: the record contains and binds a description; re-hashed here, matches). P3-probe-delta <- the clean+injected probe records (A2, observational: two RECORDED decisions, DENY vs ALLOW on the identical request). P5-detection <- full JUnit, the two named assertions (A2, scoped detection). The two CANNOT ESTABLISH have NO establishing observation: P4-violation-attribution (causal attribution of the delta to the injection — needs environment equivalence + applicable policy, not supplied) and P6-universal-coverage (universal coverage not shown; 30 other gate tests passed under this injection). Common dependency: all four observations are products of the SAME workflow run — corroboration, not mutual independence. All cited digests were re-hashed here against the frozen MANIFEST.md (match) · *ADMISSION — four direct observations from one frozen, re-hashed CI run drive six propositions; no interpretive candidate; admitted without a separate interpretive admission step* · `per-proposition detail in the sections above`

Frozen reconstruction manifest (the inputs a replay fixes) — reconstruction_id FN-da06acd59938f5b9 · observation_model FN-CUSTOS-003 hash sha256:1c8e3a182457689aeaead139abe1f5deb4f2191d0d4771dfb5b0b28536a66ee3 · observation_set hash sha256:c282f2bcbb8a49c31c7ce0b5a68819256235ccf9c6a597135628dc1817d509e5 (4 observations: o_1c108b5af3196e43, o_5997ddacd23affde, o_9dfcd958f303c9d6, o_a3d4c62abfee2417) · rule versions fn-reconstruction-engine-v1 / fn-rule-pattern-v1 / fn-composition-v1 / fn-temporal-v1 / fn-canon-v1. Internal determinism: re-running against these frozen inputs reproduces the identifier and the six proposition outcomes; the engine is not public, so independent verification is at the OBSERVATION level — the run-11 set is frozen in-repo (docs/red-proof-records/run-38610363998/, MANIFEST.md) and the cited digests were re-fetched and re-hashed here against it (match); anyone can re-walk and re-hash the same files · *FROZEN — mirrors Annex D binding* · `Annex D (binding)`

Annex A — Admitted observation set

The admitted half of the candidate admission register — each an accountable ADMIT decision (determiner and criterion below). The full register (admitted and rejected) is in the machine-readable finding.

Candidate	Observation	Primary	SHA-256	Determined by	Criterion
obs-verdict	run:38610363998 · verdict_consistent_with_declared_rule	003-weaken-action-gate/red-proof-evidence.json	53cf1126b0e3	FactNotebook	direct machine record; SHA-256 re-hashed vs MANIFEST.md (match); contributes to P1
obs-modification	run:38610363998 · record_contains_sealed_modification_description	003-weaken-action-gate/red-proof-injection-diff.patch	74456f767f1c	FactNotebook	sealed diff (SHA-256-identified byte-commitment) re-hashed vs MANIFEST.md (match); contributes to P2
obs-probe-delta	run:38610363998 · probe_delta_deny_clean_allow_injected	003-probe/red-proof-probe-{clean,injected}.json	5683de843dc4	FactNotebook	clean (b5951061…) + injected (5683de84…) both re-hashed vs MANIFEST.md (match); contributes to P3
obs-detection	run:38610363998 · regression_detected_by_identified_assertions	003-weaken-action-gate/red-proof-junit.xml	bffc84099f61	FactNotebook	JUnit (+ pytest-output eb89afc7…) re-hashed vs MANIFEST.md (match); contributes to P5

Annex B — Non-admitted candidate register

Every candidate was admitted: none rejected, none left unresolved.

Annex C — Acquisition / selection boundary

Records were constituted and supplied by Tavares Heath's CUSTOS-CORE red-proof CI (the SUBJECT; public repo ceyptoslim/CUSTOS-CORE, AGPL). The record is the subject's own CI output for a blind, pre-registered run; it is NOT a source independent of the subject, and admitted without selective exclusion of candidate observations — all extracted candidates were admitted. This finding is bounded to that submitted surface; the absence of internal selection does not establish the submission's completeness relative to the represented system, nor whether an adverse class of observations was

withheld — that completeness is not established.

Surface constituted / submitted by	Tavares Heath's CUSTOS-CORE red-proof CI (the SUBJECT; public repo ceyptoslim/CUSTOS-CORE, AGPL). The record is the subject's own CI output for a blind, pre-registered run; it is NOT a source independent of the subject
Submission form	one public CI run (38610363998 / run 11, sabotage 003): evidence record, clean+injected probe decisions, JUnit (no -x), exit-code and sealed injection-diff — now frozen in-repo at docs/red-proof-records/run-38610363998/ (MANIFEST.md) and re-hashed here (match)
Candidate extraction instrument	the subject's CI (pytest under a known-bad sabotage patch verified against a frozen DESIGN.md hash, no -x; plus a control probe on clean + injected builds). FactNotebook re-reads the preserved record; it does NOT re-run the suite or the probe
Admission authority / mechanism	FactNotebook (proposer). Acquisition + integrity are kept distinct from independence and from relevance/sufficiency. The record is the subject's own (same-workflow corroboration, not mutual independence). The sealed diff (74456f76…), both probe artifacts, JUnit, pytest output and exit code were re-hashed here against the frozen MANIFEST.md and match. Not an independent audit
Reconstruction engine	camsva 0.9.0
Surface type	submitted
Acquisition mode	submitted bundle — no candidate selection
Period	run 38610363998 (commit f5d48c9), workflow_dispatch on merged main, 2026-10-14
Known limitations	Bounded to one blind, pre-registered run (38610363998 / run 11, sabotage 003) against ONE control (the pre-flight action gate). The run-11 set is now frozen in-repo (docs/red-proof-records/run-38610363998/, MANIFEST.md); the CITED digests (the injection diff 74456f76…, both probe artifacts, JUnit, pytest output and exit code) were re-hashed here and match — digest concordance only, i.e. the retrieved bytes match the manifest used. This does NOT by itself establish that the frozen set corresponds to the original CI run's own artifacts (that would need comparison to the run's outputs), nor independence of production (the manifest is a product of the same workflow). Only the files this reconstruction cites were re-hashed, not necessarily every file in the directory. PUBLICATION IS GATED on the subject's draft review. The probe establishes the gate's DECISION, not downstream execution or an external effect. Detection is of this regression by the two identified assertions, not universal coverage. What is established is what the preserved record permits — not that CUSTOS-CORE's controls are adequate.

Annex D — Policy & engine binding

Engine camsva 0.9.0 · Reconstruction ID `FN-da06acd59938f5b9`

FactNotebook reconstructs what can be established from the surviving records — including what cannot be — and keeps the establishment separate from the judgment applied to it.